MILITARY SPECIFICATION

TRANSISTOR, TYPE 2N1046

1. SCOPE

1.1 Scope.- This specification covers the detailed requirements for a PNP diffused germanium high frequency power transistor and shall be in accordance with Specification MIL-S-19500 except as otherwise specified herein. The unit has the following characteristics:

	h_{FE} I_C =-0.5 Adc	V_{CE} (sat) I_C =-1Adc	f_T I_C = -0.5Adc	h_{fe} f = 10 mc	y_{FE}
	---	Vdc	mc	db	mhos
Minimum	40	---	15	3	1.25
Maximum	200	-1.0	---	---	

1.2 Maximum Ratings.-

P_C T_C= 25°C	T_j	T_{stg}	V_{CB}	V_{EB}	V_{CE}
W	°C	°C	Vdc	Vdc	Vdc
30 See fig. 2	+100	-55 to +100	-100	-1.5	-50

2. APPLICABLE DOCUMENTS

2.1 The following specification, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATION

MILITARY
MIL-S-19500 - Semiconductor Devices, General Specification for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Design and construction.- Transistors shall be of the design, construction and physical dimensions as shown on figure 1.

3.1.1 Operating position.- Transistors shall operate in any position.

3.2 Performance characteristics.- Performance characteristics shall be as specified in 4.3 and 4.4.

3.3 Marking.- The transistor shall be marked with the "USN" prefix in lieu of "JAN" prefix (after qualification has been obtained).

4. QUALITY ASSURANCE PROVISIONS

FSC 5960

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4.1 Qualification inspection.- Qualification inspection shall consist of examinations and tests specified in 4.3 and 4.4.

4.2 Acceptance inspection.- Acceptance inspection shall consist of examinations and tests specified under groups A and B.

4.3 Group A inspection.- Group A inspection shall consist of the examination and tests specified in table I.

4.4 Group B inspection.- Group B inspection shall consist of the examinations and tests specified in table II.

4.4.1 Disposition of sample units.- Sample units which have been subjected to and have passed subgroups 1, 5, 6 and 7 of group B inspection may be delivered on the contract or order, providing these sample units are subjected to and pass group A inspection, after group B inspection is terminated.

4.5 Life test after shipment.- In the event a sample fails the cutoff life or storage life test, and the inspection lot represented has been shipped prior to the completion of the cutoff life or storage life test, the supplier shall immediately notify the inspector and the Bureau of Ships of the failure and of the action taken.

5. PREPARATION FOR DELIVERY

5.1 See Specification MIL-S-19500.

6. NOTES

6.1 Information pertaining to qualification testing may be obtained from the Bureau of Ships, Department of the Navy, Washington 25, D. C

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Preparing activity:
Navy - Bureau of Ships
(Project 5960-N028(NAVY))

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Reference Paragraph MIL-S-19500	Examination or Test	Conditions	AQL	Symbol	Limits Min.	Limits Max.	Unit
	Subgroup 1 Visual and mechanical examination	---		---	--	--	--
	Subgroup 2 Collector current, reverse bias	V_{CE} = -40 Vdc, V_{BE} = +0.2 Vdc	1.0	I_{CEX}	--	-2	mAdc
50.6	Collector cutoff current, emitter open	V_{CB} = -40 Vdc, I_E = 0		I_{CBO}	--	-1	mAdc
50.6	Emitter cutoff current, collector open	V_{EB} = -0.5 Vdc, I_C = 0		I_{EBO}	--	-0.1	mAdc
50.6	Emitter cutoff current, collector open	V_{EB} = -1.5 Vdc, I_C = 0		I_{EBO}	--	-2.0	mAdc
50.1	Collector to base breakdown voltage, emitter open	I_C = -10 mAdc, I_E = 0		BV_{CBO}	-100	--	Vdc
50.1	Emitter to base breakdown voltage, collector open	I_E = -2 mAdc, I_C = 0		BV_{EBO}	-1.5	--	Vdc
50.1	Base to emitter voltage	I_C = -0.5 Adc, V_{CE} = -1 Vdc		V_{BE}	--	-0.4	Vdc
50.1	Collector to emitter breakdown voltage, base open	I_C = -200 mAdc, I_B = 0 (pulse method)		BV_{CEO}	-50	--	Vdc
	Subgroup 3						
50.40	Static forward current transfer ratio	I_C = -0.5 Adc, V_{CE} = -1 Vdc	1.0	h_{FE}	40	200	--
50.25	Collector to emitter saturation voltage	I_C = -1 Adc, I_B = -0.1 Adc		$V_{CE(sat)}$	--	-1	Vdc
	Subgroup 4 Small-signal short circuit forward current transfer ratio cutoff frequency (where h_{fe} is at unity)	I_C = -0.5 Adc, V_{CE} = -10 Vdc	2.5	f_T	15	--	mc
50.33	Small-signal short-circuit forward current transfer ratio	I_C = -0.5 Adc, V_{CE} = -5 Vdc, f = 10 mc		h_{fe}	3	--	db
	Static forward current transfer admittance	I_C = -0.5 Adc, V_{CE} = -2 Vdc		y_{fe}	1.35	--	mhos

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Table II - Group B inspection.

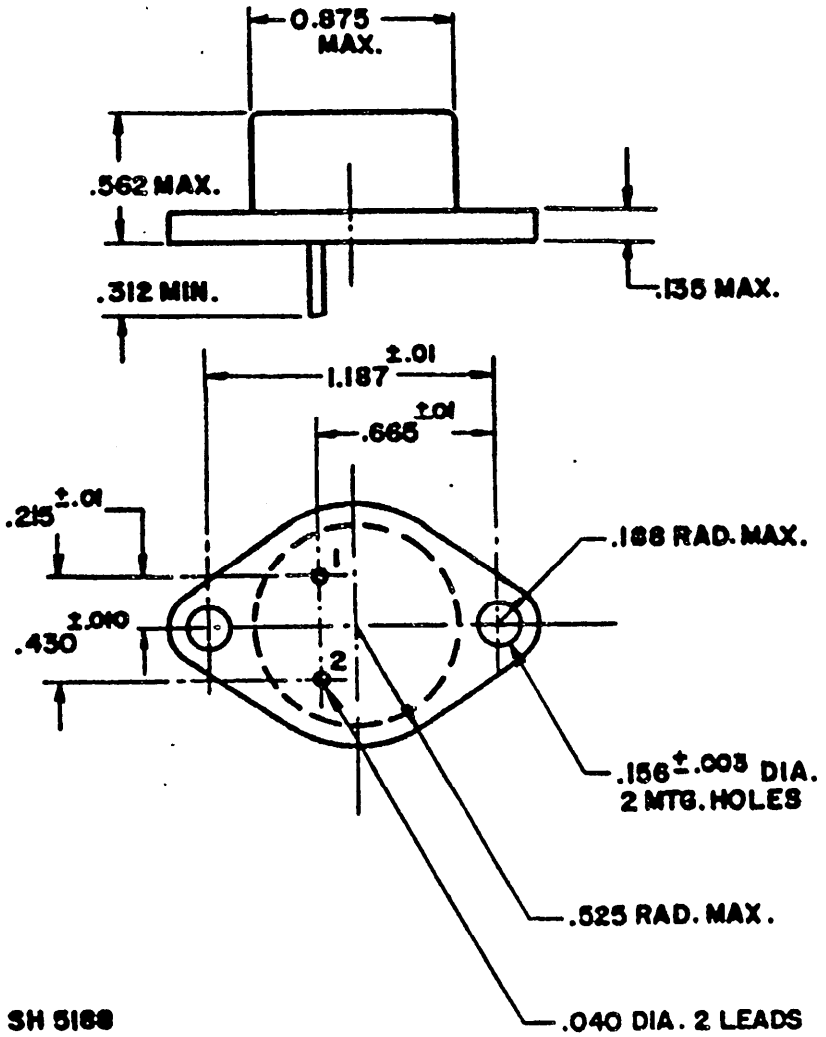
Reference Paragraph MIL-S-19500	Examination or Test	Conditions	AQL	Sample Code Letter	Symbol	Limits		Unit
						Min.	Max.	
	Subgroup 1							
30.9	Physical dimensions	---	6.5	G	---	--	--	--
	Subgroup 2							
40.13	Soldering	---			---	--	--	--
40.14	Temperature cycling	T_A = +95° ±5° C, 5 cycles			---	--	--	--
40.16	Thermal shock	T_A = +85° to 0°C			---	--	--	--
40.6	Moisture resistance		4.0	F	---	--	--	--
	End points: (Same as subgroup 7)							
	Subgroup 3	Non-Operating						
40.10	Shock	500 G, 1 msec, 3 planes: X_1, Y_1, Z_1: 5 blows/plane			---	--	--	--
40.4	Constant acceleration	5,000 G's			---	--	--	--
40.18	Vibration fatigue	Non-operating			---	--	--	
40.20	Vibration, variable frequency	---	4.0	F	---	--	--	--
	End points: (Same as subgroup 7)							
	Subgroup 4							
40.1	Barometric pressure	15 mm Hg					-2	mAdc
	Collector cutoff current, emitter open	V_{CB} = -40 Vdc, I_E = 0			I_{CBO}		-2	mAdc
30.6	High temperature operation:	T_A = +55° C min.						
	Collector cutoff current, emitter open	V_{CB} = -40 Vdc, I_E = 0			I_{CBO}	--	-16	mAdc
40.8	Salt atmosphere, corrosion		4.0	F				
	End points: (Same as subgroup 7)							
	Subgroup 5							
20.23	Thermal resistance	---	4.0	F	θ_{J-C}	--	2.5	°C/W

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Table II - Group B inspection (cont'd).

Reference Paragraph MIL-S-19500	Examination or Test	Conditions	AQL 1,000 hour life test 250 hr	500 hr	1000 hr	Inspection Level	Symbol	Limits Min.	Max.	Unit
	Subgroup 6									
40.7	Storage life	Method A					---	---	---	--
40.7.1		T_{stg} = +95°C min.					---	---	---	--
40.7.4		1,000 hour	4.0	4.0	4.0	L8				
		500 hour	2.5	2.5						
		250 hour	1.5							
	End points: (Same as subgroup 7)									
	Subgroup 7									
	Cutoff life	Method A								
		T_{stg} = +55°C min.					---	---	---	--
		V_{CE} = -40 Vdc,								
		V_{BE} = +0.2 Vdc								
	End points:	1,000 hour	4.0	4.0	4.0	L8				
		500 hour	2.5	2.5						
		250 hour	1.5							
	Collector cut-off current, emitter open	V_{CB} = -40 Vdc, I_E = 0					I_{CBO}	---	-2	mAdc
	Static forward current transfer ratio	I_C = -0.5 Adc, V_{CE} = -1 Vdc					h_{FE}	32	---	--

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NOTES: 1. All unspecified tolerance ±0.005.

2. All dimensions in inches.

3. Collector is electrically connected to mtg. case.

   Lead 1 ---- Emitter

   Lead 2 ---- Base

4. All dimensions referred to leads are for eylet holes in header only

Figure 1 Transistor, Type 2N1046

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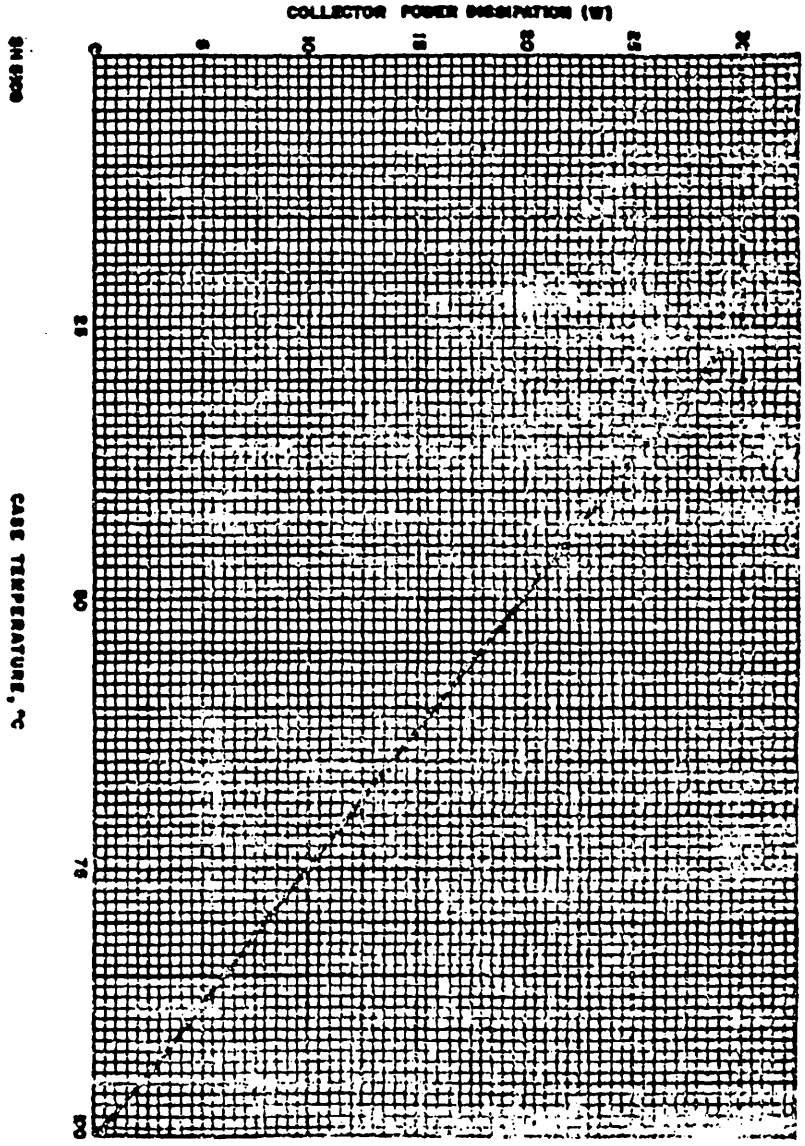


Figure 2 - Power derating curve for transistor, type 2N1613

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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